Jobsheet 1 – Konsep Dasar Pemrograman

Nama : Muhammad Fauzi Fadillah

NIM : 254107020085

Kelas : TI_1G

2.2.1. Praktikum Pemilihan

Kode program:

```
package jobsheet1;


import java.util.Scanner;


public class PraktikumPemilihan {
    public static double HitungNilai(double tugas, double kuis, double uts, double uas){
        double nilaiAkhir;


        nilaiAkhir = (tugas * 0.2) + (kuis * 0.2) + (uts * 0.3) + (uas * 0.3);


        return nilaiAkhir;
    }


    public static String HitungNilaiHuruf(double nilaiAkhir){
        if(nilaiAkhir <= 100 && nilaiAkhir > 80){
            return "A";
        }else if(nilaiAkhir <= 80 && nilaiAkhir > 73){
            return "B+";
        }else if(nilaiAkhir <= 73 && nilaiAkhir > 65){
            return "B";
        }else if(nilaiAkhir <= 65 && nilaiAkhir > 60){
            return "C+";
```

```java
    }else if(nilaiAkhir <= 60 && nilaiAkhir > 50){
        return "C";
    }else if(nilaiAkhir <= 50 && nilaiAkhir > 39){
        return "D";
    }else if(nilaiAkhir <= 39 && nilaiAkhir >= 0){
        return "E";
    }else{
        return "Tidak Valid";
    }
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    double tugas, kuis, uts, uas;
    int counter = 0;

    System.out.println("Program Menghitung Nilai Akhir");
    System.out.println("=====================");

    System.out.print("Masukkan Nilai Tugas: ");
    tugas = sc.nextDouble();

    System.out.print("Masukkan Nilai Kuis: ");
    kuis = sc.nextDouble();

    System.out.print("Masukkan Nilai UTS: ");
    uts = sc.nextDouble();

    System.out.print("Masukkan Nilai UAS: ");
```

```
uas = sc.nextDouble();

if(tugas > 100 || tugas < 0){
    counter++;
}else if(kuis > 100 || tugas < 0){
    counter++;
}else if(uts > 100 || tugas < 0){
    counter++;
}else if(uas > 100 || tugas < 0){
    counter++;
}

double nilaiAkhir = HitungNilai(tugas, kuis, uts, uas);
String nilaiHuruf = HitungNilaiHuruf(nilaiAkhir);

if(counter > 0){
    System.out.println("======================");
    System.out.println("======================");
    System.out.println("nilai tidak valid (nilai 0-100 saja)");
    System.out.println("======================");
    System.out.println("======================");
}else{
    System.out.println("======================");
    System.out.println("======================");
    System.out.println("Nilai Akhir : " + nilaiAkhir);
    System.out.println("Nilai Huruf : " + nilaiHuruf);
    System.out.println("======================");
    System.out.println("======================");
```

```
            if(nilaiAkhir > 50){
                System.out.println("SELAMAT ANDA LULUS");
            }else{
                System.out.println("MAAF ANDA TIDAK LULUS");
            }
        }
    }
}
```

Output kode program tidak valid:

```
Program Menghitung Nilai Akhir
======================
Masukkan Nilai Tugas: 85
Masukkan Nilai Kuis: 90
Masukkan Nilai UTS: 120
Masukkan Nilai UAS: 70
======================
======================
nilai tidak valid (nilai 0-100 saja)
======================
======================
```

Output kode program valid :

```
Program Menghitung Nilai Akhir
======================
Masukkan Nilai Tugas: 90
Masukkan Nilai Kuis: 100
Masukkan Nilai UTS: 70
Masukkan Nilai UAS: 90
======================
======================
Nilai Akhir : 86.0
Nilai Huruf : A
======================
======================
SELAMAT ANDA LULUS
```

2.3.1. Praktikum Perulangan

Kode program :

```
package jobsheet1;

import java.util.Scanner;

public class PraktikumPerulangan {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double nim, n;
        // int n;

        System.out.print("Masukkan NIM anda: ");
        nim = sc.nextDouble();
        n = nim % 100;

        if(n < 10){
            n += 10;
        }

        for(int i = 1; i <= n; i++) {
            if(i == 10 || i == 15){

            }else if(i % 3 == 0){
                System.out.print("# ");
            }else if(i % 2 != 0){
                System.out.print("* ");
            }else{
                System.out.print(i + " ");
```

```
            }
        }
    }
}
```

Output program n<10 :

```
Masukkan NIM anda: 254107020008
* 2 # 4 * # * 8 # * # * 14 16 * #
```

Output program n = 85 :

```
Masukkan NIM anda: 254107020085
* 2 # 4 * # * 8 # * # * 14 16 * # * 20 # 22 * # * 26 # 28 * # * 32 # 34 * # * 38 # 40 * # * 44 # 46 * # * 50 # 52 * # * 56 # 58 * # * 62 # 64 * # * 68 # 70 * # * 74 # 76 * # * 80 # 82 *
# *
```

2.4.1. Praktikum Array

Kode program :

```
package jobsheet1;

import java.util.Scanner;

public class PraktikumArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        // String[] MK = {"Pancasila", "Konsep Teknologi Informasi" + "Critical Thinking dan Problem Solving", "Matematika Dasar",
        // "Bahasa Inggris", "Dasar Pemrograman", "Praktikum Dasar Pemrograman", "Keselamatan dan Kesehatan Kerja"};
        String[] MK = new String[8];
        double[][] NilaiMK = new double[MK.length][3]; // 8 baris matkul, dan 3 kolom (nilai angka, bobot sks, nilai setara)
        String[] NilaiHuruf = new String[MK.length];
        double totalSKS = 0;
        double totalNilai = 0;
        double IP = 0;
```

```
System.out.println("=======================");
System.out.println("Program Menghitung IP Semester");
System.out.println("=======================");

for (int i = 0; i < MK.length; i++) {
    while (true) {
        System.out.print("Masukkan nama Mata Kuliah: ");
        MK[i] = sc.nextLine();

        // System.out.print("Masukkan nilai Angka untuk MK " + MK[i] + ": ");
        System.out.print("Masukkan nilai Angka: ");
        NilaiMK[i][0] = sc.nextDouble();

        System.out.print("Masukkan bobot SKS: ");
        NilaiMK[i][1] = sc.nextDouble();

        sc.nextLine();

        if(NilaiMK[i][0] <= 100 && NilaiMK[i][0] >= 0){
            break;
        }else{
            System.out.println("Input tidak valid. Masukkan nilai (0 - 100)");
        }
    }

    if(NilaiMK[i][0] <= 100 && NilaiMK[i][0] > 80){
        NilaiMK[i][2] = 4; //
        NilaiHuruf[i] = "A";
```

```
    }else if(NilaiMK[i][0] <= 80 && NilaiMK[i][0] > 73){
        NilaiMK[i][2] = 3.5;
        NilaiHuruf[i] = "B+";
    }else if(NilaiMK[i][0] <= 73 && NilaiMK[i][0] > 65){
        NilaiMK[i][2] = 3;
        NilaiHuruf[i] = "B";
    }else if(NilaiMK[i][0] <= 65 && NilaiMK[i][0] > 60){
        NilaiMK[i][2] = 2.5;
        NilaiHuruf[i] = "C+";
    }else if(NilaiMK[i][0] <= 60 && NilaiMK[i][0] > 50){
        NilaiMK[i][2] = 2;
        NilaiHuruf[i] = "C";
    }else if(NilaiMK[i][0] <= 50 && NilaiMK[i][0] > 39){
        NilaiMK[i][2] = 1;
        NilaiHuruf[i] = "D";
    }else if(NilaiMK[i][0] <= 39 && NilaiMK[i][0] >= 0){
        NilaiMK[i][2] = 0;
        NilaiHuruf[i] = "E";
    }else{
        NilaiHuruf[i] = "Tidak Valid";
    }
}

for (int i = 0; i < MK.length; i++) {
    totalSKS += NilaiMK[i][1];
    totalNilai += NilaiMK[i][1] * NilaiMK[i][2]; // bobot sks * nilai setara
}

IP = totalNilai / totalSKS;
```

```
        System.out.println("======================");
        System.out.println("Hasil Konversi Nilai");
        System.out.println("======================");

        System.out.printf("%-60s %-12s %-12s %-12s%n",
                "Mata Kuliah", "Nilai Angka", "Nilai Huruf", "Bobot Nilai");

        for (int i = 0; i < MK.length; i++) {
            System.out.printf("%-60s %-12.2f %-12s %-12.2f%n",
                    MK[i], NilaiMK[i][0], NilaiHuruf[i], NilaiMK[i][2]);
        }
        System.out.println("======================");
        System.out.println("IP:\t" + IP);
    }
}
```

Output program:

```
======================
Program Menghitung IP Semester
======================
Masukkan nama Mata Kuliah: Pancasila
Masukkan nilai Angka: 75
Masukkan bobot SKS: 2
Masukkan nama Mata Kuliah: Konsep Teknolog Informasi
Masukkan nilai Angka: 85
Masukkan bobot SKS: 4
Masukkan nama Mata Kuliah: Critical Thinking and Problem Solving
Masukkan nilai Angka: 70
Masukkan bobot SKS: 4
Masukkan nama Mata Kuliah: Matematika Dasar
Masukkan nilai Angka: 85
Masukkan bobot SKS: 4
Masukkan nama Mata Kuliah: Bahasa Inggris
Masukkan nilai Angka: 85
Masukkan bobot SKS: 4
Masukkan nama Mata Kuliah: Dasar Pemrograman
Masukkan nilai Angka: 62
Masukkan bobot SKS: 4
Masukkan nama Mata Kuliah: Praktikum Dasar Pemrograman
Masukkan nilai Angka: 62
Masukkan bobot SKS: 6
Masukkan nama Mata Kuliah: Keselamatan dan Kesehatan Kerja
Masukkan nilai Angka: 85
Masukkan bobot SKS: 4
======================
Hasil Konversi Nilai
======================
Mata Kuliah                                                 Nilai Angka  Nilai Huruf  Bobot Nilai
Pancasila                                                   75.00        B+           3.50
Konsep Teknolog Informasi                                   85.00        A            4.00
Critical Thinking and Problem Solving                       70.00        B            3.00
Matematika Dasar                                            85.00        A            4.00
Bahasa Inggris                                              85.00        A            4.00
Dasar Pemrograman                                           62.00        C+           2.50
Praktikum Dasar Pemrograman                                 62.00        C+           2.50
Keselamatan dan Kesehatan Kerja                             85.00        A            4.00
======================
IP:     3.375
```

2.5.1. Praktikum Fungsi

Kode program :

```
package jobsheet1;

public class PraktikumFungsi {

   static String[] bunga = {"Aglonema", "Keladi", "Alocasia", "Mawar"};

   static int jumlahCabang = 4;

   static int[][] stok = {

         {10, 5, 15, 7},

         {6, 11, 9, 12},

         {2, 10, 10, 5},
```

```
        {5, 7, 12, 9}
    };

  static double[] hargaBunga = {75_000, 50_000, 60_000, 10_000};

  public static void tampilPendapatan(){
    double[] pendapatanCabang = new double[jumlahCabang];

    for (int i = 0; i < jumlahCabang; i++) {
      double penjualanAglonema = stok[i][0] * hargaBunga[0];
      double penjualanKeladi = stok[i][1] * hargaBunga[1];
      double penjualanAlocasia = stok[i][2] * hargaBunga[2];
      double penjualanMawar = stok[i][3] * hargaBunga[3];

      pendapatanCabang[i] = penjualanAglonema + penjualanKeladi + penjualanAlocasia +
penjualanMawar;
    }

    for (int i = 0; i < jumlahCabang; i++) {
      System.out.print("RoyalGarden" + (i+1) + ": Rp." + pendapatanCabang[i]);
      if(pendapatanCabang[i] > 1_500_000){
        System.out.println(" (Status: Sangat Baik)");
      }else{
        System.out.println(" (Status: Perlu Evaluasi)");
      }
    }
  }
  public static void main(String[] args) {
    tampilPendapatan();
```

```
    }
}
```

Output program :

```
RoyalGarden1: Rp.1970000.0 (Status: Sangat Baik)
RoyalGarden2: Rp.1660000.0 (Status: Sangat Baik)
RoyalGarden3: Rp.1300000.0 (Status: Perlu Evaluasi)
RoyalGarden4: Rp.1535000.0 (Status: Sangat Baik)
PS C:\College\semester2\ASD\Praktikum\jobsheet1>
```

Tugas 1

Kode program :

```
package jobsheet1;

import java.util.Scanner;

public class Tugas1 {
    private static char[] KODE = {'A', 'B', 'D', 'E', 'F', 'G', 'H', 'L', 'N', 'T'};
    public static char[][] KOTA = {
        {'B','A','N','T','E','N'},
        {'J','A','K','A','R','T','A'},
        {'B','A','N','D','U','N','G'},
        {'C','I','R','E','B','O','N'},
        {'B','O','G','O','R'},
        {'P','E','K','A','L','O','N','G','A','N'},
        {'S','E','M','A','R','A','N','G'},
        {'S','U','R','A','B','A','Y','A'},
        {'M','A','L','A','N','G'},
        {'T','E','G','A','L'}
    };
```

```java
    public static int cariIndexPlat(char plat){
        int indexPlat = 0;
        for (int i = 0; i < KODE.length; i++) {
            if(plat == KODE[i]){
                indexPlat = i;
            }
        }

        return indexPlat;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Masukkan plat nomor: ");
        char inputPlat = sc.next().charAt(0);

        int indexPlat = cariIndexPlat(inputPlat);

        for (int i = 0; i < KOTA[indexPlat].length; i++) {
            if(KOTA[i] != null){
                System.out.print(KOTA[indexPlat][i]);
            }else{
                break;
            }
        }
    }
}
```

Ouput program :

```
Masukkan plat nomor: N
MALANG
```

Tugas 2

Kode program :

```
package jobsheet1;


import java.util.Scanner;


public class Tugas2 {
    static Scanner sc = new Scanner(System.in);


    public static void main(String[] args) {
        System.out.println("=== PROGRAM MANAJEMEN JADWAL KULIAH ===");


        System.out.print("Masukkan jumlah mata kuliah yang ingin didaftarkan: ");
        int n = sc.nextInt();
        sc.nextLine();


        String[][] jadwal = new String[n][4];


        inputJadwal(jadwal, n);


        boolean running = true;
        while (running) {
            System.out.println("\n--- MENU ---");
            System.out.println("1. Tampilkan Seluruh Jadwal");
```

```java
System.out.println("2. Cari Jadwal Berdasarkan Hari");
System.out.println("3. Cari Jadwal Berdasarkan Nama Mata Kuliah");
System.out.println("4. Keluar");
System.out.print("Pilih menu (1-4): ");
int pilihan = sc.nextInt();
sc.nextLine();

switch (pilihan) {
  case 1:
    tampilkanSeluruhJadwal(jadwal, n);
    break;
  case 2:
    System.out.print("Masukkan hari yang ingin dicari (misal: Senin): ");
    String hariCari = sc.nextLine();
    tampilkanJadwalPerHari(jadwal, n, hariCari);
    break;
  case 3:
    // d. Menampilkan berdasarkan nama matkul
    System.out.print("Masukkan nama mata kuliah yang dicari: ");
    String matkulCari = sc.nextLine();
    cariMataKuliah(jadwal, n, matkulCari);
    break;
  case 4:
    System.out.println("Terima kasih, program selesai.");
    running = false;
    break;
  default:
    System.out.println("Pilihan tidak valid!");
}
```

```
        }
    }

    static void inputJadwal(String[][] data, int n) {
        System.out.println("\n--- INPUT DATA JADWAL ---");
        for (int i = 0; i < n; i++) {
            System.out.println("Data ke-" + (i + 1));

            System.out.print("Nama Mata Kuliah : ");
            data[i][0] = sc.nextLine();

            System.out.print("Ruang            : ");
            data[i][1] = sc.nextLine();

            System.out.print("Hari             : ");
            data[i][2] = sc.nextLine();

            System.out.print("Jam (mis: 08:00) : ");
            data[i][3] = sc.nextLine();

            System.out.println("-------------------------");
        }
    }

    static void tampilkanSeluruhJadwal(String[][] data, int n) {
        System.out.println("\n--- DAFTAR SELURUH JADWAL KULIAH ---");
        System.out.printf("%-25s %-15s %-10s %-15s\n", "Mata Kuliah", "Ruang", "Hari",
"Jam");
        System.out.println("-----------------------------------------------------------------");
```

```
    for (int i = 0; i < n; i++) {
        System.out.printf("%-25s %-15s %-10s %-15s\n",
            data[i][0],
            data[i][1],
            data[i][2],
            data[i][3]
        );
    }
    System.out.println("----------------------------------------------------------------------");
}

static void tampilkanJadwalPerHari(String[][] data, int n, String hari) {
    System.out.println("\n--- JADWAL HARI: " + hari.toUpperCase() + " ---");
    boolean ditemukan = false;

    System.out.printf("%-25s %-15s %-15s\n", "Mata Kuliah", "Ruang", "Jam");
    System.out.println("----------------------------------------------------------");

    for (int i = 0; i < n; i++) {
        if (data[i][2].equalsIgnoreCase(hari)) {
            System.out.printf("%-25s %-15s %-15s\n", data[i][0], data[i][1], data[i][3]);
            ditemukan = true;
        }
    }

    if (!ditemukan) {
        System.out.println("Tidak ada jadwal kuliah pada hari " + hari);
    }
```

```java
        System.out.println("------------------------------------------------------------------");
    }

    static void cariMataKuliah(String[][] data, int n, String namaMatkul) {
        System.out.println("\n--- PENCARIAN MATA KULIAH: " + namaMatkul + " ---");
        boolean ditemukan = false;

        System.out.printf("%-25s %-15s %-10s %-15s\n", "Mata Kuliah", "Ruang", "Hari",
"Jam");
        System.out.println("---------------------------------------------------------------------------------");

        for (int i = 0; i < n; i++) {
            if (data[i][0].toLowerCase().contains(namaMatkul.toLowerCase())) {
                System.out.printf("%-25s %-15s %-10s %-15s\n",
                    data[i][0], data[i][1], data[i][2], data[i][3]
                );
                ditemukan = true;
            }
        }

        if (!ditemukan) {
            System.out.println("Mata kuliah \"" + namaMatkul + "\" tidak ditemukan.");
        }
        System.out.println("---------------------------------------------------------------------------------");
    }
}
```

Inputan program :

```
=== PROGRAM MANAJEMEN JADWAL KULIAH ===
Masukkan jumlah mata kuliah yang ingin didaftarkan: 5

--- INPUT DATA JADWAL ---
Data ke-1
Nama Mata Kuliah : Pancasila
Ruang            : RT09_8T
Hari             : Rabu
Jam (mis: 08:00) : 07:00 - 08:40
-------------------------
Data ke-2
Nama Mata Kuliah : Dasar Pemrograman
Ruang            : RT04_5B
Hari             : Senin
Jam (mis: 08:00) : 10:30 - 14:30
-------------------------
Data ke-3
Nama Mata Kuliah : Matematika Dasar
Ruang            : LPY2_6T
Hari             : Selasa
Jam (mis: 08:00) : 14:30 - 18:00
-------------------------
Data ke-4
Nama Mata Kuliah : Bahasa Inggris
Ruang            : RT01_5B
Hari             : Kamis
Jam (mis: 08:00) : 10:30 - 14:30
-------------------------
Data ke-5
Nama Mata Kuliah : Konsep Teknologi Informasi
Ruang            : LPY1_5B
Hari             : Jumat
Jam (mis: 08:00) : 10:30 - 14:30
-------------------------
```

Output menu 1 :

```
--- MENU ---
1. Tampilkan Seluruh Jadwal
2. Cari Jadwal Berdasarkan Hari
3. Cari Jadwal Berdasarkan Nama Mata Kuliah
4. Keluar
Pilih menu (1-4): 1

--- DAFTAR SELURUH JADWAL KULIAH ---
Mata Kuliah                Ruang           Hari       Jam
-------------------------------------------------------------------
Pancasila                  RT09_8T         Rabu       07:00 - 08:40
Dasar Pemrograman          RT04_5B         Senin      10:30 - 14:30
Matematika Dasar           LPY2_6T         Selasa     14:30 - 18:00
Bahasa Inggris             RT01_5B         Kamis      10:30 - 14:30
Konsep Teknologi Informasi LPY1_5B          Jumat      10:30 - 14:30
-------------------------------------------------------------------
```

Output menu 2 :

```
--- MENU ---
1. Tampilkan Seluruh Jadwal
2. Cari Jadwal Berdasarkan Hari
3. Cari Jadwal Berdasarkan Nama Mata Kuliah
4. Keluar
Pilih menu (1-4): 2
Masukkan hari yang ingin dicari (misal: Senin): Kamis

--- JADWAL HARI: KAMIS ---
Mata Kuliah                Ruang           Jam
-------------------------------------------------------
Bahasa Inggris             RT01_5B         10:30 - 14:30
-------------------------------------------------------
```

Output menu 3 :

```
--- MENU ---
1. Tampilkan Seluruh Jadwal
2. Cari Jadwal Berdasarkan Hari
3. Cari Jadwal Berdasarkan Nama Mata Kuliah
4. Keluar
Pilih menu (1-4): 3
Masukkan nama mata kuliah yang dicari: Matematika Dasar

--- PENCARIAN MATA KULIAH: Matematika Dasar ---
Mata Kuliah               Ruang           Hari       Jam
-------------------------------------------------------------------
Matematika Dasar          LPY2_6T         Selasa     14:30 - 18:00
-------------------------------------------------------------------
```